

The best three options

Build option B. Twelve VL53L8CX in a ring at 200 mm, six VL53L4CD cliff sensors at 150 mm, one RPLIDAR C1 at 100 mm, and eight AMG8833 thermal arrays at 250 mm. USD 909, about 1.0 A at 5 V. It is the cheapest build that answers all four questions the robot has to ask, and every part of it is in stock today.

	A - Collision only	B - Collision + animate	C - Full classification
Time-of-flight modules	18	19	24
People sensors	0	8 thermal	6 thermal + 4 cameras
Cost	USD 455	USD 909	USD 1,440
Average current at 5 V	724 mA	999 mA	3.8 A
Compute needed	RP2040	RP2040	Pi 5 + Hailo-8L
Will I hit it?	Yes, 360 deg	Yes, plus dark targets and sunbeams	Yes
Is it alive?	No	Yes, ~2.4 m pet / ~5 m human	Yes, in darkness too
Human or pet?	No	By fusion, to about 5 ft	Labelled, to 10 ft
Sees a black cat on dark carpet	To 1.55 m	To 6 m	To 6 m
Sees a sleeping cat	Only as an obstacle	Yes, as an animal	Yes

Option A - collision-ring, USD 455

Twelve VL53L8CX multizone carriers (Pololu 3419, USD 24.95) in a ring at 200 mm on 30 deg spacing, six VL53L4CD (Pololu 3692, USD 13.95) canted 23 deg down at 150 mm for cliffs, two PCA9548 multiplexers, and a compliant foam bumper.

What it buys. A bumper made of light. Full 360 deg obstacle detection with range, closing its blind wedges 338 mm past the skin, refreshing fast enough to stop a 0.5 m/s robot.

What it cannot do. Anything in the brief except collision avoidance. It has no human channel and no pet channel, and it loses a dark target past about 1.55 m in a lit room. In a patch of direct sun it reports *empty space*, not noise, which is the dangerous failure.

Option B - collision-ring + animate-layer, USD 909 - RECOMMENDED

Option A plus one Slamtec RPLIDAR C1 (USD 69) with its scan plane at 100 mm, eight AMG8833 thermal arrays (Adafruit 3538, USD 44.95) level at 250 mm on 45 deg spacing, and two RCWL-1601 ultrasonic rangefinders forward for glass.

Why the C1 is the best USD 69 in the study. It fixes both structural weaknesses of a ToF ring with one cable: 6 m against a 10 percent black target where the ring manages 1.55 m, a 40,000 lux rating where the ring is only characterised to 5 kLux, a plane low enough to cut a cat lying on its side, and 500 points per revolution - enough to scan-match a background map, after which anything that does not fit the map is dynamic. That is the strongest animate cue available without a camera.

Why thermal and not radar. Thermal is the only modality that separates animate from inanimate, and the only one that sees a *motionless* target. A sleeping cat is invisible to motion-gated radar and invisible to PIR, and obvious to thermal.

What it cannot do. Cat versus dog. Classification in a dark room beyond thermal's pixel limit. Anything at all in direct sunlight.

Option C - full-classification-stack, USD 1,440

Sixteen VL53L8CX, six MLX90640 32x24 thermal arrays, a VL53L8CH histogram head, the C1, four wide-angle cameras and a Raspberry Pi 5 with a Hailo-8L.

What the extra USD 531 buys. Labelled classification out to 10 ft, and a ring tight enough that the blind wedge closes 168 mm past the skin - inside the robot's own footprint - which lets the compliant bumper stop being part of the safety argument.

Why it is third. It costs 2.8 A more and needs a Pi 5 to run. One of its six MLX90640 arrays is out of stock as of 2026-09-12, and two of its line items have no published price. Buy it only if labelled classification is a requirement rather than a wish.

How to choose

Pick **A** if the robot only has to not hit things and you will keep it away from people and animals. Pick **B** in every other case. Pick **C** only if you need the robot to say the word "person" or "pet" out loud and act differently on each, at ten feet, reliably.

The gap that matters is A to B: USD 454 and 275 mA buys the entire animate layer plus the two fixes - dark targets and sunlit floors - that would otherwise put the robot on the floor or on the cat. The gap from B to C buys one capability for USD 531 and a different computer.

What your three robots should actually do

None of the three projects in this repository can take Option B as drawn. Their electronics, not their mechanics, are the binding constraint.

Project	What fits	Cost	Deciding constraint
dalek	2 x SEN0610 radar, 1 x VL53L4CX	USD 40.75	0.30 A spare and zero free GPIO. Its bumper cannot be cut, so radar behind the 1.8 mm skirt is the only sensor that needs no aperture.
r2d2	1 x VL53L8CX, 1 x SEN0610	USD 37.85	47 g of mass margin against a 9 kg ceiling, and one free pin.
fable-r2d2	9 x VL53L8CX, 1 x MLX90640, 1 x STHS34PF80	USD 314.45	Nothing binds - a Pi 4, 4.1 A spare and an entirely unused STEMMA QT bus. Its body STLs are not drawn yet, so the apertures can still be designed in.

Build the full Option B on fable-r2d2. It is the only one with the current, the bus and the unwritten CAD to absorb it.

Four things no option does

- **Direct sunlight.** ST publishes to 5 kLux and the C1 to 40 kLux. Direct sun is 100 to 120 kLux. No number exists beyond that; do not extrapolate one.
- **Mirrors.** A mirror at 45 deg returns a folded path and the robot believes the corridor continues. Map them as permanent keep-out zones.
- **Cat versus dog,** without a camera.
- **Any safety rating.** A certified IEC 61496 Type 3 scanner must detect an 1.8 percent reflectance target in its protective field. A VL53L8CX does not meet that. Use the industrial arithmetic; do not claim the rating.

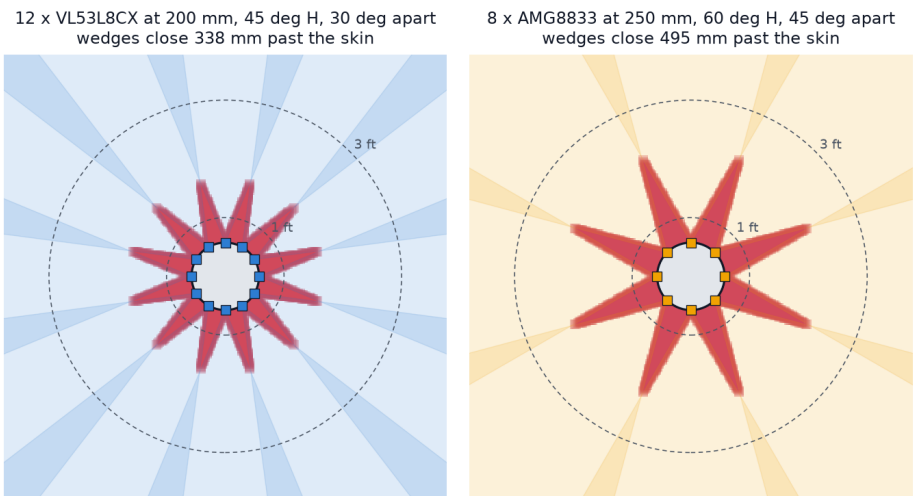
Three decisions worth more than the parts list, and they are free

1. **Make the platform round.** A 350 mm square has a 72.487 mm corner protrusion that four 45 deg sensors on the face centres leave permanently uncovered.
2. **Cap speed at 0.5 m/s.** A 15 kg robot at 1.0 m/s with a rigid shell delivers about 1500 N into a shin against a 130 N ISO/TS 15066 limit. At 0.5 m/s with a 50 mm foam bumper it delivers 38 N. Speed is the cheapest safety component in the build.
3. **Never let classification gate the stop.** There are 167 to 323 ms between the warning field and the protective field. Thermal needs 250 ms for one frame. Brake on raw geometry at frame rate; classify in parallel and change behaviour only after the robot has stopped.

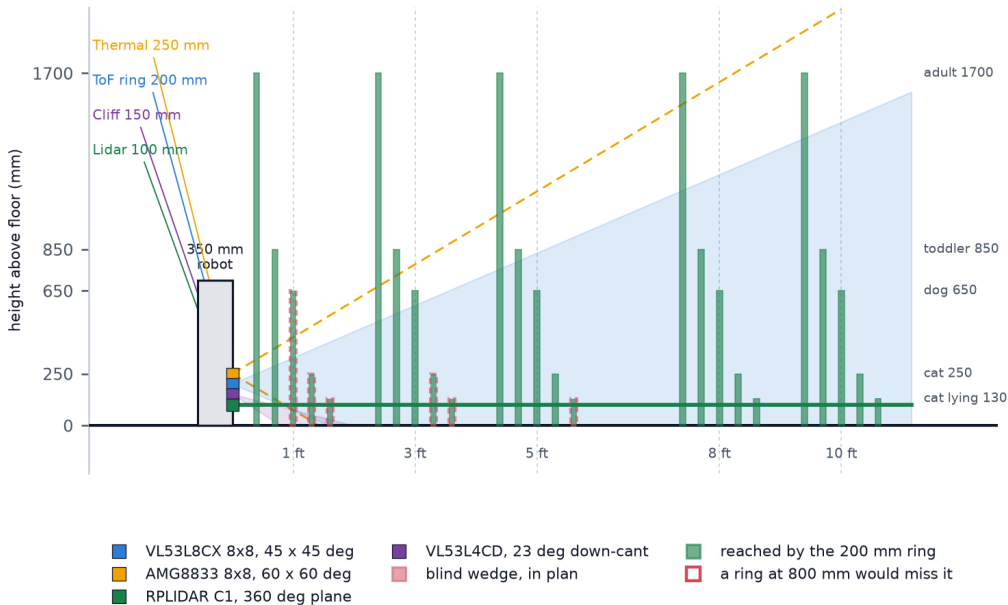
Before ordering either radar, measure the C4001 supply current. DFRobot does not publish it, and it is the one number that can still move the dalek and r2d2 totals. That is a one-meter measurement, not a research problem. Prices read 2026-09-12; nothing has been purchased or physically tested.

Coverage map: Option B on a 350 mm round platform

Plan views are evaluated point by point rather than assumed: red is ground no cone reaches. Both rings close, and the near-field wedges they leave are what the compliant bumper is in the bill of materials for. The elevation is the argument for the 200 mm mounting height - at 800 mm the same ring sails over a cat and a lying cat at close range, which is precisely what a robot runs over.



Elevation: why the ring goes at 200 mm. Solid green = the 200 mm ring reaches it. Red dashed outline = the same ring at body height (800 mm) would miss it entirely.



Every distance ring, cone angle, wedge-closure figure and subject height in this drawing is produced by radar-compare/scripts/geometry.py, the same code that generates the tables in the full reference, so the drawing cannot disagree with them.